

Project Initialization and Planning Phase

Date	03 July 2026
Team ID	
Project Title	Credit Card Approval Prediction
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposed solution aims to automate the **credit card approval process** using machine learning, improving the speed, accuracy, and consistency of approval decisions. The system analyzes applicant financial and demographic information to predict whether a credit card application should be approved or rejected. By reducing manual intervention, it helps banks minimize processing time, reduce human errors, and improve customer satisfaction. The project integrates the best-performing machine learning model into a Flask web application and supports IBM Watson Machine Learning cloud deployment for real-time predictions.

Project Overview	
Objective	The primary objective is to develop an intelligent Credit Card Approval Prediction System using machine learning techniques to provide fast, accurate, and reliable approval decisions based on applicant information.
Scope	The project covers data preprocessing, feature engineering, model training, evaluation using multiple classification algorithms (Logistic Regression, Decision Tree, Random Forest, and XGBoost), Flask web application development, and IBM Watson Machine Learning cloud deployment.
Problem Statement	
Description	Banks receive thousands of credit card applications every day. Manual evaluation is time-consuming, prone to human error, and may result in inconsistent approval decisions, leading to operational inefficiencies and customer dissatisfaction.

Impact	Automating the approval process improves decision accuracy, reduces processing time, minimizes financial risk, enhances operational efficiency, and provides a better customer experience.
Proposed Solution	
Approach	Develop a machine learning-based prediction system that analyzes applicant financial and demographic information to determine whether a credit card application should be approved or rejected. The best-performing model is deployed through a Flask web application and IBM Watson Machine Learning for real-time predictions.
Key Features	1. Machine learning-based credit card approval prediction. 2. Comparison of Logistic Regression, Decision Tree, Random Forest, and XGBoost models. 3. Automatic data preprocessing and feature engineering. 4. Real-time prediction using a Flask web application. 5. IBM Watson Machine Learning cloud deployment. 6. Fast, accurate, and user-friendly prediction interface.



Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	Computing Resources	intel Core i5/Ryzen 5 processor or higher
Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	256 GB SSD or above

Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	scikit-learn, XGBoost, pandas, NumPy, matplotlib, joblib
Development Environment	IDE	Jupyter Notebook, pycharm or vs code
Data		
Data	Source, size, format	Credit Card Approval Dataset (CSV format) from Kaggle/UCI Repository